Separable equations — 4.3

Definition: A separable differential equation is any equation of the form: $y' = f(x)\,g(y)$.

Examples: $y' = y\sin(x)$, $y' = \ln(y)(x^3+2)$

$y' = xy + 3x - 2y - 6 = (x-2)(y+3)$

$y' = f(x)$, $y' = ky$

How do we solve such equations?

$\rightarrow$ check for constant solutions

For other solutions, we "separate" variables:

Rewrite $y' = f(x)g(y)$ as $\frac{1}{g(y)} y' = f(x)$

and then integrate $\int \frac{1}{g(y)} y' \, dx = \int f(x) \, dx$

Substitute $u = y(x)$ $\quad \int \frac{1}{g(u)} \, du = \int f(x) dx$

$du = y'(x) dx$

$$\frac{dy}{dx} = f(x) g(y) \iff dy = f(x) g(y) \, dx$$

$$\iff \frac{1}{g(y)} dy = f(x) dx \iff \boxed{\int \frac{1}{g(y)} dy = \int f(x) dx}$$

<u>Example</u>: Find a general solution to $y' = (x^2-4)(3y+2)$.

→ Constant solution: $y = \frac{-2}{3}$

$$\int \frac{1}{3y+2}\,dy = \int x^2-4\,dx$$

$$\frac{1}{3}\ln|3y+2| = \frac{x^3}{3} - 4x + C$$

$$\ln|3y+2| = x^3 - 12x + C$$

$$|3y+2| = e^{x^3-12x+C} = e^{x^3-12x}e^{C} = Ce^{x^3-12x}, \quad C>0$$

$$3y+2 = Ce^{x^3-12x}$$

$$y = \frac{Ce^{x^3-12x}-2}{3} = \boxed{Ce^{x^3-12x} - \frac{2}{3}}$$

Example: Solve the IVP $y' = \frac{x}{y}$, $y(0) = 2$.

$$\rightarrow \quad \int y\,dy = \int x\,dx \quad \Rightarrow \quad \frac{y^2}{2} = \frac{x^2}{2} + C$$

$$y^2 = x^2 + C \quad \Rightarrow \quad y = \pm\sqrt{x^2 + C}$$

Solving for C: $2 = y(0) = \pm\sqrt{0+C} = \pm\sqrt{C}$, $C = 4$

$$y = \sqrt{x^2 + 4}$$

Example: Solve $y' = \frac{3x+\cos(x)}{2y+y^2}$ with $y(0)=3$.

$$\int 2y+y^2 \, dy = \int 3x+\cos(x) \, dx$$

$$y^2 + \frac{y^3}{3} = \frac{3x^2}{2} + \sin(x) + C$$

hard to solve for y in terms of x

$\longrightarrow$ leave answer implicit

Can still solve for C: $x=0$, $y=3$

$$9+9 = 0+0+C \implies C=18$$

$$y^2 + \frac{y^3}{3} = \frac{3x^2}{2} + \sin(x) + 18.$$